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Temporal trends in the use of oral anticoagulants and clinical outcomes in older, elderly, and very elderly patients with atrial fibrillation: a Danish nationwide study

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Received 23 January 2025; revised 29 April 2025; accepted 12 October 2025; online publish-ahead-of-print 18 November 2025

See the editorial comment for this article ‘Anticoagulation safety for stroke prevention in the very elderly: lessons from a Danish nationwide study’, by B. Rocca and L. Guasti <https://doi.org/10.1093/eurheartj/ehaf1017>.

Abstract

Background and Aims

Anticoagulation therapy in patients with atrial fibrillation (AF) has changed over time, particularly following the introduction of direct oral anticoagulants. However, it is unknown how the uptake of anticoagulation therapy and the related clinical outcomes in elderly patients with AF have changed over time.

Methods

Patients with new-onset AF were included and divided into three age groups: older adults (65–74 years), elderly (75–84 years), and very elderly (≥85 years). Temporal trends in the initiation of oral anticoagulants (OACs), the stroke-free survival, major bleedings including intracerebral haemorrhage (ICH), and all-cause mortality were investigated from 1999 to 2022.

Results

In total, 243 938 patients were included, of whom 89 184 (36.6%) were older adults, 99 002 were elderly (40.6%), and 55 752 (22.8%) were very elderly. The proportion of very elderly patients with AF receiving OACs was 71% in 2022. The absolute improvement in the 5-year probability of stroke-free survival was 10.1% in the older adult patients, 12.8% in the elderly patients, and 3.5% in the very elderly patients. The 5-year absolute risk of ICH increased among the elderly and very elderly AF patients.

Conclusions

Over the past two decades, the risk of stroke decreased significantly among all age groups, with no subsequent increase in the risk of bleeding among older adults in an era where OACs were almost fully implemented. In the very elderly patients aged ≥85 years, the risk of stroke only slightly improved over time, with an increase in the risk of ICH.

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† Joint first authorship: Casper Binding and Mariam Elmegaard contributed equally to the planning of the study, manuscript preparation, conduction of analysis, and revision of the manuscript.
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Structured Graphical Abstract

Key Question

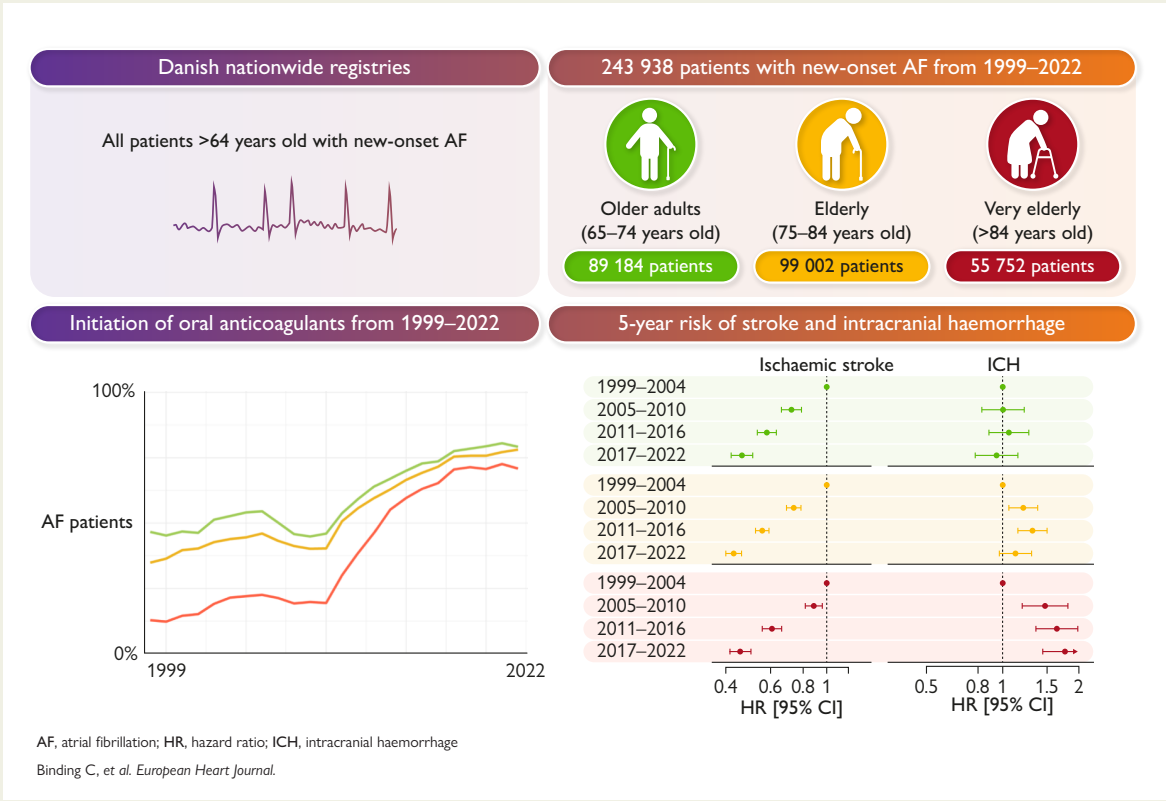
How did initiation of oral anticoagulant (OAC) treatment and clinically related outcomes in older adult, elderly, and very elderly patients with atrial fibrillation (AF) develop from 1999 to 2022?

Key Finding

In this Danish nationwide study, OACs were fully implemented in AF patients during the last two decades, which was associated to decreasing risk of stroke. Very elderly patients experienced an increased risk of intracranial haemorrhage.

Take Home Message

Current risk stratification may be inadequate in very elderly AF patients.



Study design and results. Abbreviations: AF, Atrial fibrillation; HR, Hazard ratio; ICH, Intracerebral haemorrhage.

Keywords Atrial fibrillation • Elderly • Epidemiology • Oral anticoagulants

Introduction

Atrial fibrillation (AF) is a common cardiac arrhythmia among elderly patients and is associated with a high risk of ischaemic stroke.¹ The number of patients with AF worldwide continues to rise, and due to demographic changes, patients tend to be older at the time of diagnosis which is associated with an increased risk of stroke.^{2–5} Overall, this has led to a heightened focus on stroke prevention because stroke poses a major threat towards the quality of life in the affected individuals. Furthermore, the long and complex hospital admissions related to stroke are also a significant burden to the healthcare system.^{6–8} Until 2010, vitamin K antagonists (VKAs) and acetylsalicylic acid (ASA) were considered equivalent in stroke prevention among patients with AF, despite ASA’s questionable efficacy.^{9,10} In 2010, the European

Guidelines for stroke prevention in AF patients were changed following the large randomized clinical trials (RCTs) of direct oral anticoagulants (DOACs).^{11–15} As a result, VKAs were largely replaced by DOACs, and ASA was removed from the guidelines.¹⁶ However, the clinical trials supporting the current guideline recommendations had an under-representation of the very elderly patients (≥85 years). Few smaller register-based studies investigated the topic among patients >90 years, indicating increased bleeding events among these patients,^{17,18} yet information on the safety and efficacy and prescription rates of treatment with DOACs among these patients is still lacking. Therefore, we investigated temporal trends in the initiation and withdrawal of oral anticoagulants (OACs), the risk of stroke, the stroke-free survival, the risk of major bleedings, and the risk of death among older adults, elderly, and very elderly patients with AF, from 1999 to 2022.

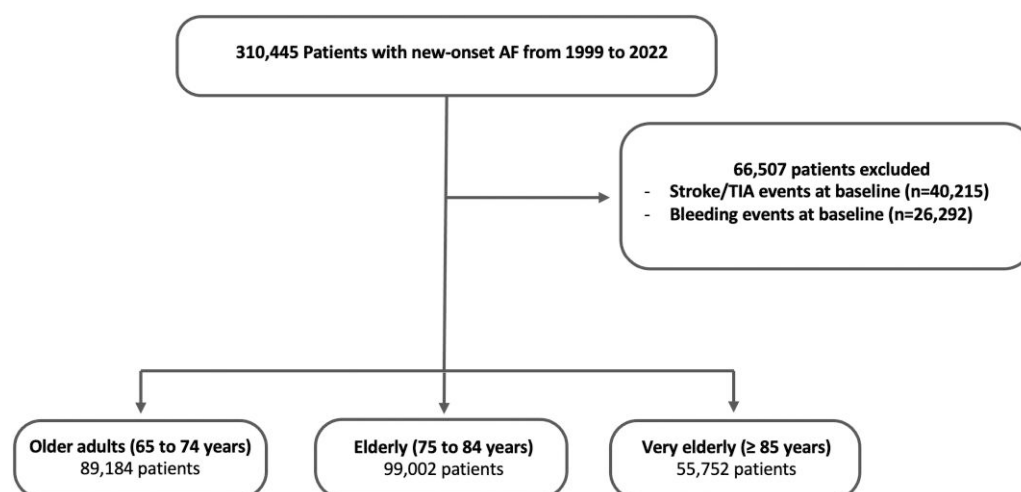


Figure 1 Study flowchart. Patients with new-onset AF from 1999 to 2022 were included and separated into three age groups based on age at index

Methods

Study design

The structure of this observational study follows the principles described in the STROBE statement.¹⁹ Data for this study were provided by Statistics Denmark.²⁰ We obtained information from four different registers, including 1) the Danish Civil Registration System,²¹ 2) the Danish National Prescription Registry,²² 3) the Danish National Patient Registry,²³ and 4) the Danish Register of Causes of Death.²⁴ Information on sex, date of birth, and emigration was obtained from the Danish Civil Registration system. All hospital contacts and their corresponding diagnoses were obtained from the Danish National Patient Registry. Diagnoses were classified according to the International Classification of Diseases, 10th edition (ICD-10). Information on all prescriptions redeemed at Danish pharmacies was retrieved from the Danish National Prescription Registry. Prescription data were classified according to the Anatomical Therapeutic Chemical (ATC) classification system. Date and cause of death were obtained from the Danish Register of Causes of Death.

Study population and exposures

We identified all patients ≥ 65 years with new-onset AF between 1 January 1999, and 31 December 2022. We included both inpatients and outpatients with either a primary or secondary diagnosis of AF. Patients with prior bleedings, prior ischaemic stroke, and prior transient ischaemic attack (TIA) were excluded from this study. The index date was set as the day of the first registered AF diagnosis for outpatients and as the day of discharge for patients hospitalized with a first registered AF diagnosis. In terms of exposures, patients were divided into three age groups according to their age at index: 1) older adults (65–74); 2) elderly (75–84 years); and 3) very elderly (≥ 85 years). Furthermore, patients were divided into four calendar year groups: 1999–2004, 2005–10, 2011–16, and 2017–22, according to the year of inclusion. Comorbidities were defined from registered ICD-10 diagnoses five years leading up to inclusion, and concomitant medications were defined from prescriptions redeemed within six months up to inclusion. Prior OAC treatment was defined from a prescription within 5 years prior to the index. Study flowchart is shown in [Figure 1](#), and a visual representation of the study design is provided in [Supplementary data online, Figure S1](#).

Outcomes and follow-up

OAC and ASA initiation was estimated from redemption of prescriptions for DOACs, VKAs, and ASA at AF diagnosis. The main outcomes of interest

in this study were the risk of ischaemic stroke, the stroke-free survival, and the risk of death. Stroke-free survival referred to the timespan in which a person was alive and remained free from experiencing an ischaemic stroke. Using stroke-free survival as an outcome handles the challenge associated with the high risk of death in the very elderly patients with AF. The definition of ischaemic stroke included all ischaemic strokes (ICD-10: I63) and unspecified strokes (ICD-10: I64) recorded as a primary or secondary diagnosis in either an outpatient or inpatient setting.²⁵ Death was defined as death from all causes. Other outcomes of interest included major bleeding,²⁶ fatal bleeding, and intracerebral haemorrhage (ICH).²⁷ Major bleeding was defined as an inpatient hospital admission with either a primary or secondary diagnosis of gastrointestinal (GI) bleeding, airways bleeding, urinary-tract bleeding, or ICH. Fatal bleeding and fatal stroke were defined from causes of death if either was listed as the primary or secondary cause of death. ICH included all major bleedings with a registered intracerebral bleeding diagnosis. GI bleedings were defined from inpatient admissions with diagnosis codes of bleeding in the GI system. ICD-10 codes used to define baseline characteristics and outcomes are provided in [Supplementary data online, Tables S1–3](#), whereas ATC codes used to define concomitant medications are provided in [Supplementary data online, Table S4](#). Each patient was followed until the occurrence of an outcome, death, emigration, the end of the study period (31 December 2022), or for a maximum follow-up of 5 years.

Statistical analyses

Baseline characteristics were reported as counts followed by percentages or median followed by interquartile range (IQR). We performed a temporal trend analysis using four calendar year groups to evaluate changes in outcomes over a 24-year study period. The Kaplan-Meier estimator was used to calculate the 5-year cumulative incidence of all-cause mortality and fatal bleeding. The Aalen-Johansen estimator²⁸ was used to calculate the 5-year cumulative incidence of ischaemic stroke, stroke-free survival, major bleeding, and ICH, accounting for the competing risk of death and censoring.

We used cause-specific Cox models²⁹ to estimate the hazard ratios (HR) of ischaemic stroke, death, and the composite endpoint of ischaemic stroke and death across the four calendar groups in intention-to-treat analyses. The assumptions for using the model (lack of interaction, proportional hazard, and linearity) were fulfilled unless otherwise stated. All models were adjusted for pre-defined potential confounders such as sex, age at index, CHA₂DS₂-VA at baseline, antiplatelet drug therapy at baseline, and chronic kidney disease at baseline. Additionally, we adjusted for abnormal liver function at baseline, non-steroidal anti-inflammatory drugs (NSAIDs) at

baseline, and alcohol abuse at baseline in models evaluating fatal bleeding, major bleeding, and ICH. The first calendar year group (1999–2004) was used as the reference group in all analyses. HRs with 95% confidence intervals (CI) were presented in forest plots alongside the absolute risks in each calendar year group. Additionally, we calculated the restricted mean survival time (RMST) for stroke-free survival between the first (1999–2004) and the latest calendar year group (2017–22) to illustrate the number of days gained without ischaemic stroke in both age groups during the follow-up period.

We estimated the yearly proportions of patients with new-onset AF who were started on OACs or ASA in the time interval from 180 days before to 90 days after the index. This was done to include the patients who were initiated in OAC or ASA in primary care, before being referred to an outpatient clinic where an AF diagnosis was registered. Patients who died within the first 90 days from the index were not part of this proportion evaluation. In all risk analyses, patients were included on the day of AF diagnosis, emigration was a censoring condition, and death served as a competing risk.

We also investigated the trend in proton pump inhibitor (PPI) usage at the time of index. This was done by evaluating the proportion of patients who claimed a prescription of PPIs between the index and the following 90 days.

We performed interaction analyses to investigate how age group (older adults, elderly, and very elderly) influenced the risk of outcome according to calendar year in the primary and secondary analyses. *P*-values < .05 indicated an interaction with age.

Data management and analyses were performed in SAS³⁰ (version 9.04.01), and in R Studio³¹ (version 2022.07.0.548).

Supplemental analyses

As a supplemental analysis, we calculated the 5-year cumulative incidence of fatal stroke, non-fatal stroke, non-fatal bleeding, and GI bleedings (ICD-10 definitions in [supplementary data online, Table S2](#)). Furthermore, we evaluated the risk of primary and secondary outcomes during 2 years of follow-up, to investigate whether observed associations were present at an early stage.

Results

Patient characteristics

In total, 243 938 patients aged 65 years or older with new-onset AF were included and divided into three age groups (65–74 years, 75–84 years, and ≥85 years) and four calendar year groups, each covering a period of 6 years as follows: 47 296 (19.4%) patients diagnosed with new-onset AF in 1999–2004, 51 924 (21.3%) patients in 2005–10, 69 492 (28.5%) patients in 2011–16, and 75 226 (30.8%) patients in 2017–22. The overall median age of the included patients was 78 years (IQR: 72–84 years), and 49.6% were male. The median CHA₂DS₂-VA among the included patients was 2.5 (SD: 1) across all four calendar year groups. Baseline characteristics of the patients stratified by calendar year are presented in [Table 1](#). Additionally, baseline characteristics stratified according to the three age groups are provided in [Supplementary data online, Table S5](#).

Temporal trend in initiation of OAC and ASA from 1999 to 2022

[Figure 2A](#) shows the OAC prescription rates between 1999 to 2022. Among older adults, OAC initiation remained stable between 1999 to 2012, with initiation percentages around 46%. Initiation increased slightly from 35% to 40% among elderly patients and from 13% to 19% among very elderly patients. From 2010 to 2018, we observed a steady increase in the yearly proportion of patients initiating OAC

therapy in older adults, elderly, and very elderly patients, and by 2022, both older adults and elderly patients reached initiation percentages of 79% while initiation percentage among very elderly patients reached 71%. Among older adults and elderly patients initiated on OAC, 55% and 57% received DOAC, and 45% and 43% received VKAs, respectively. Among very elderly patients receiving OAC, 69% were prescribed DOACs, and 31% were prescribed VKAs. For ASA, prescription rates remained stable with small increases in the period from 1999 to 2010; however, we observed a gradual decline in the prescription of ASA from around 20% in 2010 to very few initiations in 2022 in all three groups ([Figure 2B](#)).

Temporal trend in stroke-free survival and ischaemic stroke from 1999 to 2022

The 5-year probability of stroke-free survival increased from 65.9% (95% CI: 65.1%–66.6%) in the first calendar year group to 76.0% (95% CI: 75.2%–76.7%) in the latest calendar year group among older adults, and from 45.6% (95% CI: 45.0%–46.3%) to 58.4% (95% CI: 57.6%–59.2%) among elderly patients ([Figure 3A](#)). Among the very elderly patients, the absolute five-year probability of stroke-free survival increased from 20.6% (95% CI: 19.9%–21.4%) in the first calendar year group to 24.1% (95% CI: 23.1%–25.2%) in the latest calendar year group. The absolute 5-year stroke-free survival was improved by 10.1% in older adults, by 12.8% in the elderly, and by 3.5% in the very elderly from 1999 to 2022 ([Figure 3B](#)). The corresponding RMSTs during the 5-year follow-up period were estimated to 123 (95% CI: 119–127) among older adults, 160 days (95% CI: 156–164) among the elderly patients, and 96 days (95% CI: 90–102) among the very elderly patients. The absolute 5-year risk of ischaemic stroke decreased significantly from 7.2% (95% CI: 6.8%–7.6%) in the first calendar year group to 3.8% (95% CI: 3.5%–4.1%) in the latest calendar year group among the older adults (see [Supplementary data online, Figure S2A](#)). Among elderly patients, the absolute 5-year risk of ischaemic stroke likewise decreased from 10.6% (95% CI: 10.1%–11.0%) to from the earliest to the latest calendar year group (see [supplementary data online, Figure S2B](#)). In the very elderly patients, the 5-year absolute risk of ischaemic stroke decreased significantly from 10.4% (95% CI: 9.8%–10.9%) in the first calendar year group to 5.1% (95% CI: 4.7–5.6) in the latest calendar year group (see [Supplementary data online, Figure S2C](#)). When adjusting for potential confounders, we found a decreased risk of ischaemic stroke among older adults (HR: 0.48, 95% CI: 0.43–0.52), among elderly patients (HR: 0.42, 95% CI: 0.39–0.45), and among the very elderly patients (HR: 0.46, 95% CI: 0.42–0.51) in the most recent calendar year group compared to patients from the first calendar year group ([Figure 4](#)). Interaction analyses indicated that the association between calendar year group and both stroke-free survival and death depended on the age group of patients. This indicated that very elderly patients benefited less from the implementation of DOACs, in terms of risk of death and improvement in stroke-free survival, than the older adults and the elderly ([Figure 4](#)).

Temporal trend in fatal bleeding, major bleeding, ICH, and GI bleeding from 1999 to 2022

From the earliest calendar year group to the latest, we observed small decreases in the incidence of fatal bleedings (earliest: 1.0%, 95% CI 0.9%–1.2%, latest: 0.8%, 95% CI 0.6%–0.9%) and of major bleedings (earliest: 9.0%, 95% CI 8.6%–9.5%, latest: 7.1%, 95% CI 6.7%–7.5%)

Table 1 Baseline characteristics for AF patients stratified according to calendar year groups

	1999–2004	2005–2010	2011–2016	2017–2022
Patients (n)	47 296	51 924	69 492	75 226
Sex, male (n)	22 275 (47.1)	24 624 (47.4)	34 956 (50.3)	39 248 (52.2)
Age (y)—median [IQR]	78 [72, 84]	78 [72, 84]	77 [71, 84]	77 [72, 83]
Comorbidities, n (%)				
Ischaemic heart disease (IHD)	13 631 (28.8)	13 164 (25.4)	14 477 (20.8)	11 251 (15.0)
Hypertension	18 503 (39.1)	25 781 (49.7)	36 361 (52.3)	37 804 (50.3)
Chronic kidney failure	667 (1.4)	1300 (2.5)	2587 (3.7)	2709 (3.6)
Chronic heart failure (CHF)	11 668 (24.7)	10 110 (19.5)	10 811 (15.6)	9353 (12.4)
Peripheral vascular disease (PVD)	2730 (5.8)	3408 (6.6)	4749 (6.8)	4745 (6.3)
Abnormal liver function	310 (0.7)	394 (0.8)	759 (1.1)	928 (1.2)
Alcohol abuse	502 (1.1)	819 (1.6)	1409 (2.0)	1334 (1.8)
Chronic obstructive lung disease (COLD)	5260 (11.1)	6415 (12.4)	8321 (12.0)	6819 (9.1)
Cancer	4726 (10.0)	6344 (12.2)	10 457 (15.0)	12 409 (16.5)
Diabetes	4989 (10.5)	6539 (12.6)	10 653 (15.3)	11 591 (15.4)
CHA ₂ DS ₂ -VA—median [IQR]	2 [2,3]	2 [2,3]	2 [2,3]	2 [2,3]
Medication, n (%)				
Beta blocker (BB)	10 926 (23.1)	17 013 (32.8)	20 047 (34.1)	21 122 (35.3)
Calcium channel blocker (CCB)	10 155 (21.5)	12 888 (24.8)	16 532 (28.1)	17 232 (28.8)
Renin–angiotensin system inhibitors (RASi)	11 246 (23.8)	19 329 (37.2)	25 303 (43.1)	27 206 (45.5)
Thiazide	9607 (20.3)	11 619 (22.4)	10 453 (17.8)	8 326 (13.9)
Loop diuretic	14 242 (30.1)	14 041 (27.0)	12 352 (21.0)	10 884 (18.2)
Digoxin	11 057 (23.4)	6716 (12.9)	2 656 (4.5)	1 690 (2.8)
Statin	3728 (7.9)	12 819 (24.7)	19 243 (32.7)	21 439 (35.8)
Non-steroid anti-inflammatory drugs (NSAIDs)	9476 (20.0)	9030 (17.4)	9 083 (15.5)	7 123 (11.9)
Prior OAC treatment	3317 (7.0)	5344 (10.3)	4 289 (7.3)	18 830 (31.5)
Antiplatelet drugs	10 950 (23.2)	18 904 (36.4)	22 783 (38.8)	18 324 (30.6)
Social				
Nursing home	1414 (3.0)	2013 (3.9)	3 447 (5.9)	3 210 (5.4)
Highest level of education				
Primary school or high school	34 785 (73.5)	32 251 (62.1)	29 393 (50.0)	24 093 (40.3)
Short- cycle higher Education or vocational school	8775 (18.6)	13 519 (26.0)	19 713 (33.5)	23 145 (38.7)
Bachelor's degree or Bachelor of Applied Science	2654 (5.6)	4564 (8.8)	7 413 (12.6)	9 550 (16.0)
Master's degree or PhD	1082 (2.3)	1590 (3.1)	2 255 (3.8)	3 035 (5.1)
Ethnicity				
Danish	45 861 (97.0)	50 163 (96.6)	56 618 (96.3)	57 514 (96.1)
Western	1382 (2.9)	1681 (3.2)	2074 (3.5)	2216 (3.7)
Non-Western	53 (0.1)	80 (0.2)	82 (0.1)	93 (0.2)

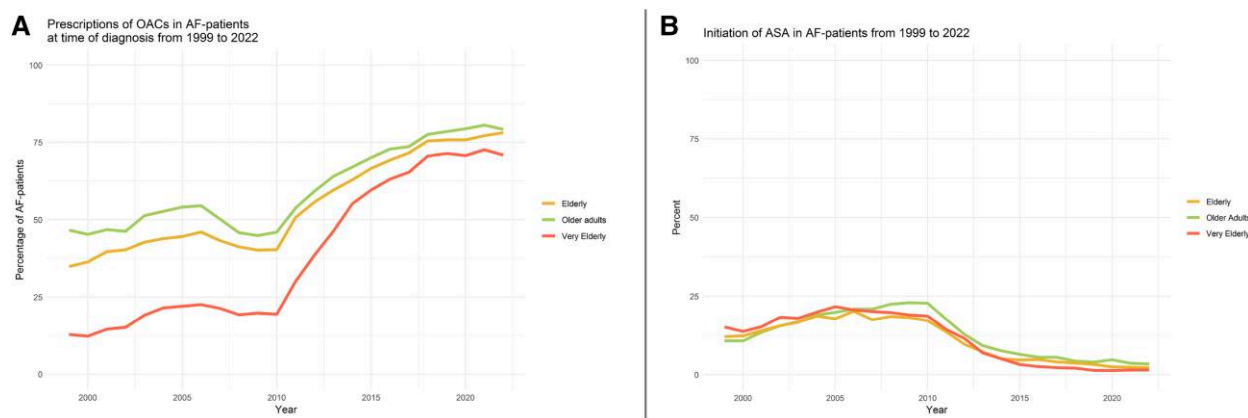


Figure 2 Initiation of OACs and Aspirin in AF patients from 1999 to 2022 stratified according to age groups

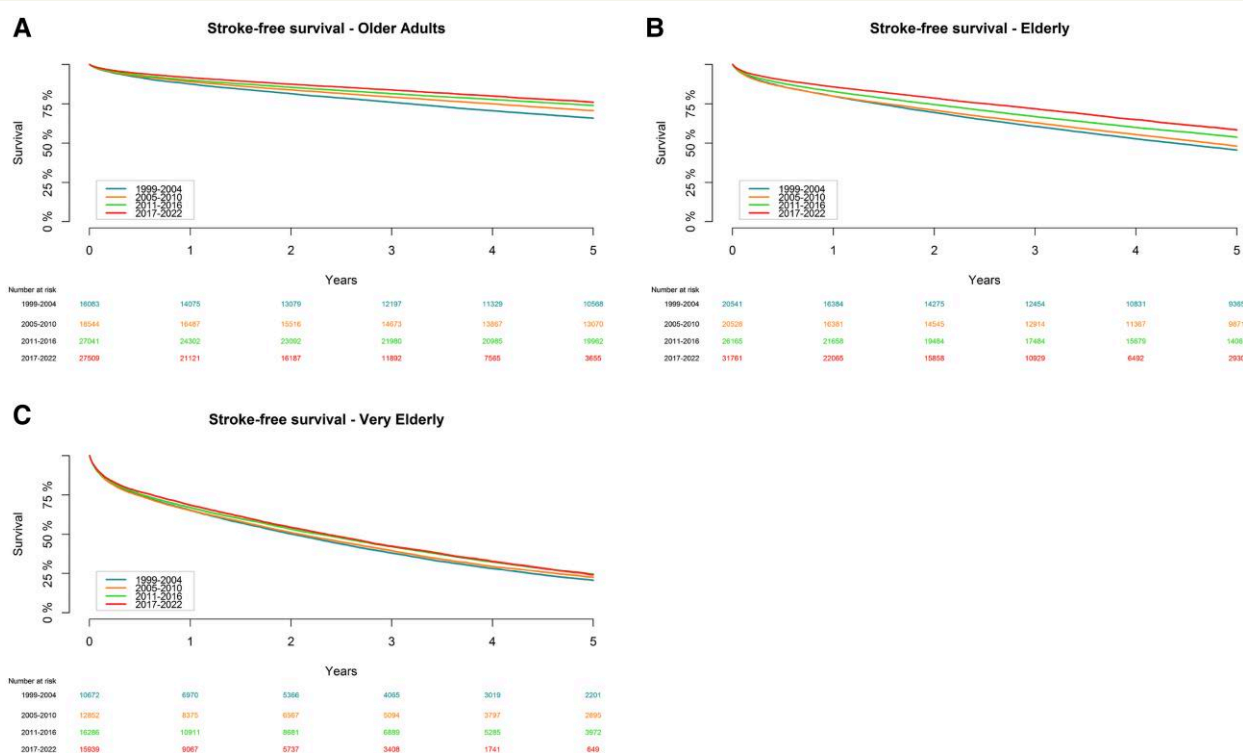


Figure 3 The 5-year stroke-free survival among older adults (A), elderly (B), and very elderly (C) AF patients from 1999 to 2022

among older adults. We observed a likewise small decline of fatal bleedings (earliest: 1.9%, 95% CI 1.6%–2.1%, latest: 1.4%, 95% CI 1.2%–1.6%) and of major bleedings (earliest: 11.7%, 95% CI 11.3%–12.2%, latest: 10.1%, 95% CI 9.6%–10.6%) among elderly patients. No such changes were observed in the very elderly patients (Figure 5, Supplementary data online, Figures S3 and S4). When focusing solely on ICH, our data, however, suggested an increased risk among the elderly and the very elderly AF patients in the most recent calendar year group. The 5-year cumulative incidence of ICH increased significantly from 1.8% (95% CI: 1.6%–2.0%) in 1999 to 2.3% (95% CI: 2.1%–2.5%) in

2022 among elderly, and from 1.3% (95% CI: 1.1%–1.5%) in 1999 to 2.5% (95% CI: 2.2%–2.8%) among the very elderly in 2022 (see Supplementary data online, Figure S5). When adjusting for potential confounders, we found a statistically significant increase in relative risk of ICH (HR: 1.55, 95% CI: 1.24–1.95) among the very elderly patients in the most recent calendar year group compared to patients from the first calendar year group (Figure 5C). Interaction analyses indicated that very elderly patients experienced less improvement in risk of major bleeding and more risk of ICH compared to older adults and elderly patients during implementation of DOACs (Figure 5). The relative

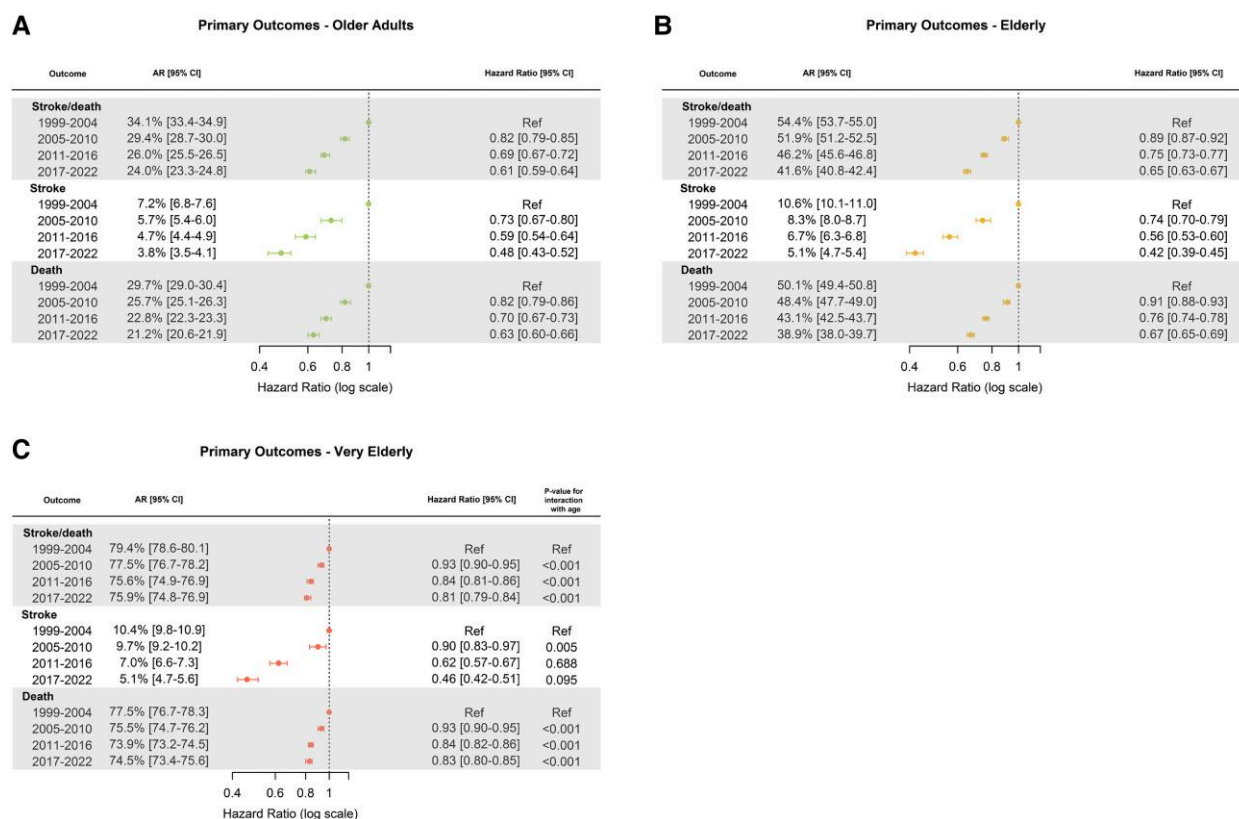


Figure 4 Forest plot for the three primary outcomes stratified according to calendar year groups: (A) the older adults, (B) the elderly, and (C) the very elderly patients. The first calendar year group served as the reference in all three primary outcome analyses

risks of GI bleedings decreased significantly from 1999 to 2022 among older adults (HR: 0.65, 95%CI: 0.58–0.72), among the elderly (HR: 0.65, 95%CI: 0.60–0.71), and among very elderly (HR: 0.71, 95%CI: 0.63–0.79) (Figure 5 and Supplementary data online, Figure S6).

Temporal trend in all-cause mortality from 1999 to 2022

In the older adult AF patients, the 5-year cumulative incidence of all-cause mortality decreased significantly from 29.7% (95% CI: 29.0%–30.4%) in the first calendar year group to 21.2% (95% CI: 20.6%–21.9%) in the latest calendar year group (see Supplementary data online, Figure S7A), resulting in an absolute risk reduction in all-cause mortality of 8.5%. Among the elderly AF patients, the 5-year absolute mortality decreased by 11.2% from 50.1% (95% CI: 49.4%–50.8%) in the first calendar year group to 38.9% (95% CI: 38.0%–39.7%) in the latest calendar year group (see Supplementary data online, Figure S7B). Among the very elderly patients, the five-year cumulative incidence of all-cause mortality decreased significantly from 77.5% (95% CI: 76.7%–78.3%) in the first calendar year group to 74.5% (95% CI: 73.8%–75.6%) in the latest calendar year group (see Supplementary data online, Figure S7C), which corresponds to an absolute risk reduction in all-cause mortality of 3%. In the adjusted models, the most recent calendar year group was associated with a lower risk of all-cause mortality compared to the first calendar year group among older adults (HR: 0.63, 95% CI: 0.60–0.66), among elderly (HR: 0.67,

95% CI: 0.65–0.69), and among very elderly patients (HR: 0.83, 95% CI: 0.80–0.85) (Figure 4).

Supplemental analyses

As shown in Supplementary data online, Table S5, the number of older adult AF patients, elderly AF patients, and very elderly AF patients increased during the 24-year study period. Only one-third of the very elderly AF patients were males. We found that the very elderly AF patients were more likely to be diagnosed with chronic heart failure compared to the older adults and the elderly patients. Furthermore, being elderly or very elderly was also associated with a higher CHA₂DS₂-VA score at baseline and a lower educational level, compared to older adults. The very elderly patients were also less likely to be prescribed other medications (except diuretics, digoxin, and antiplatelet drugs) compared to the other age groups. Both non-fatal stroke and fatal stroke decreased significantly in all three age groups throughout the study period (see Supplementary data online, Figures S8.1 and S8.2). Non-fatal bleeding decreased significantly among older adult and elderly patients but remained constant among the very elderly patients from 1999 to 2022 (see Supplementary data online, Figure S9). PPI usage at time of AF diagnosis increased from 1999 to 2010 and remained stable from 2010 to 2022 in all age groups (see Supplementary data online, Figure 10). Risk estimation of primary and secondary outcomes in a two-year follow-up period showed similar results as those obtained in the main analyses (see Supplementary data online, Figures S11 and S12).

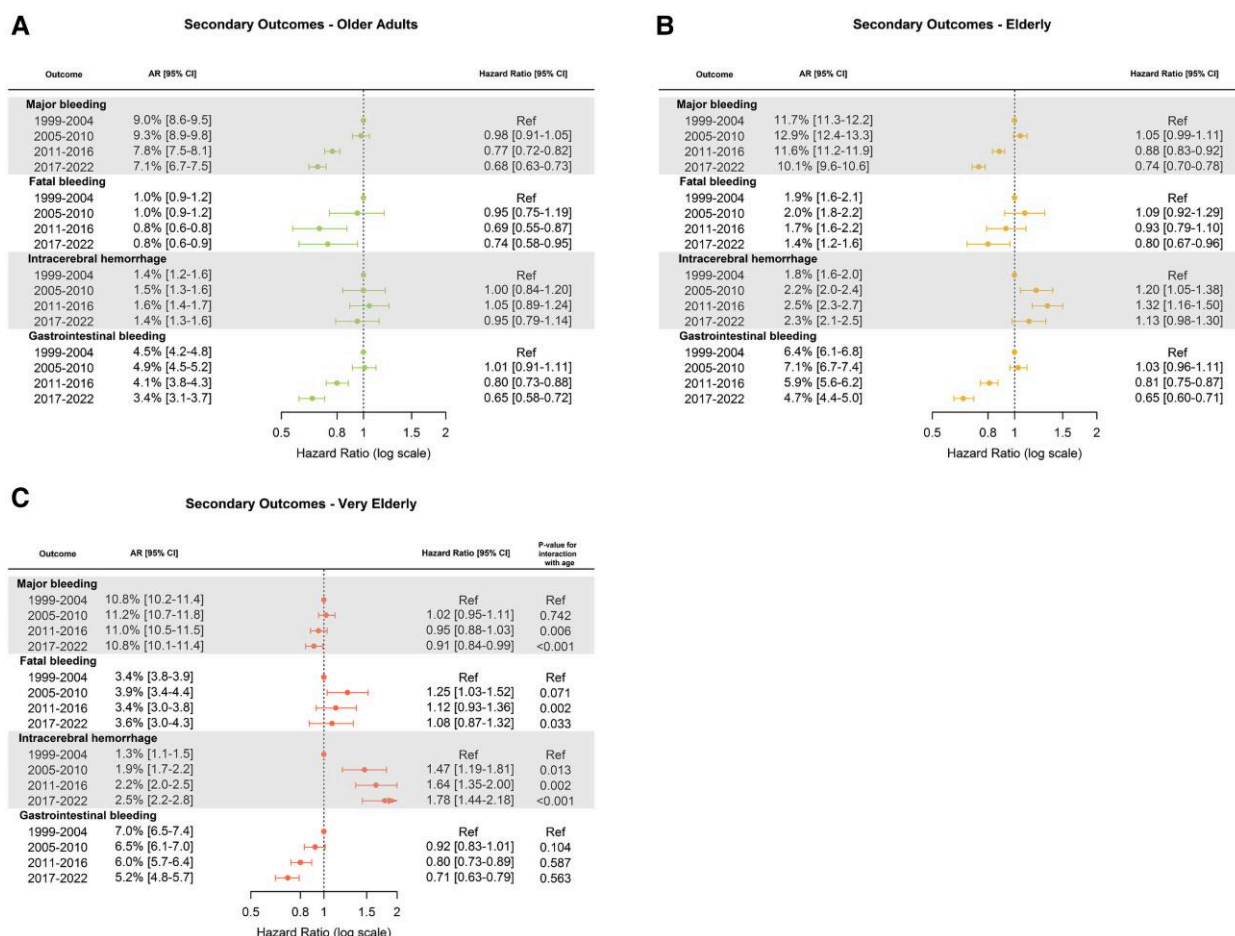


Figure 5 Forest plot for the three secondary outcomes stratified according to calendar year groups. *Figure 5A* represents the older adults, *Figure 5B* represents the elderly, and *Figure 5C* represents the very elderly patients. The first calendar year group served as the reference in all three secondary outcome analyses

Discussion

Main findings

In this Danish nationwide cohort study, we assessed temporal trends of OAC initiation and the related clinical outcomes in nationwide data of 243 938 patients with new-onset AF from 1999 to 2022, and revealed the following key findings: (i) OACs were equally prescribed among older adults, elderly and very elderly AF patients at proportions of approximately 71% to 79% in 2022, and prescription of aspirin decreased in the same period; (ii) the 5-year stroke-free survival improved by 10.1% in the older adult patients from 1999 to 2022, 12.8% in the elderly patients from 1999 to 2022, and 3.5% in the very elderly patients; and (iii) the 5-year risk of ICH nearly doubled among the very elderly patients from 1999 to 2022, although the absolute risk was low compared to the decreased absolute risk of stroke (*Structured Graphical Abstract*).

OAC prescriptions in the very elderly patients with AF

In recent years, more focus has been on the growing 'very elderly' population (aged 85 years or older), due to overall improvement in

health care, improved disease prevention strategies, and healthier lifestyles.^{32,33} The very elderly patients with AF constitute around 18% of the current European AF population, and by 2060, the total European AF population is projected to more than double.³ This challenges the decision-making process regarding OAC initiation since evidence on safety and efficacy is lacking for these patients. Despite lacking evidence on safety and efficacy among the very elderly AF patients, our results show that DOACs are widely prescribed among these patients in Denmark. The use of DOACs in the very elderly patients is poorly documented, but several studies have investigated the use of DOACs among younger patients. A Danish cohort study from 2022 examined OAC prescription rates in around 40 000 AF patients aged 75 years or older after their first AF-related hospital contact from 2010 to 2018. They found an OAC prescription rate among the included patients of 92% in 2018.³⁴ Another study conducted on data from the Get With The Guidelines-AFIB registry showed rates of around 95% in AF patients with a median age of 73 years.³⁵ Our data found OAC prescription rates at around 71% among the very elderly AF patients in 2022, which indicates lower treatment rates among very elderly patients compared to younger cohorts.

Stroke-free survival and bleeding risks associated with the use of OACs in the very elderly AF patients

The safety and efficacy of treatment with OACs among the very elderly AF patients are uncertain, as these individuals are rarely included in clinical trials investigating AF,^{11–14} and are underrepresented in RCTs in general.³⁶ Information on the safety and efficacy of OACs in very elderly AF patients mainly arises from observational studies from which it is difficult to acquire true treatment effects (both efficacy and safety) due to a lack of randomization. A subgroup analysis from the PREFER in AF trial concluded that the benefits of OACs by far outweigh the risk of bleeding in AF patients aged 85 years or older.³⁷ Another study from Taiwan investigated the risk of ischaemic stroke, ICH, major bleeding, and mortality among AF patients aged 90 years or older, and they found that treatment with DOACs was associated with a lower risk of the clinical endpoints compared to VKA and no OAC therapy, thereby supporting the use of DOACs in the very elderly patients.³⁸ In 2023, a large clinical trial (FRAIL-AF) was started to determine if it was favourable to switch elderly FRAIL AF patients from VKAs to DOACs. However, the trial was stopped after 163 primary outcomes, as the initial results demonstrated a 69% increased risk of bleeding in those individuals who were switched from VKAs to DOACs.³⁹ A significant limitation in previous studies conducted on the very elderly AF patients is the limited number of included patients. In our study, we investigated 55 752 very elderly patients with new-onset AF. Among these patients, we found an improvement in the 5-year stroke-free survival along with an increased risk of ICH over a 24-year period. The underlying cause of this trend is unclear since many factors have evolved over the past 24 years. With that in mind, it is reasonable to believe that the changes in the OAC prescription patterns over time could be associated with the increased risk of ICH among the very elderly patients observed in this study. However, future clinical trials are needed to clarify the safety of OACs among the very elderly patients.

Risk stratification of very elderly AF patients

Risk stratification among AF patients can be a complicated task, as many factors contribute to the overall stroke risk. The CHA₂DS₂-VA score is the current risk stratification tool used in the stroke risk assessment among AF patients, as it combines the most common stroke risk factors into a numeric score ranging from 0 to 8.⁴⁰ A score ≥ 2 points suggests a need for OAC therapy in order to prevent stroke. It is well established that high age strongly correlates with the risk of stroke, and according to CHA₂DS₂-VA, age ≥ 75 alone provides enough points for OAC therapy in patients with AF.^{41,42} Our results indicate that among very elderly patients in the most recent calendar year group, there is an elevated risk of ICH compared to patients from earlier calendar year groups. Altogether, this calls for improved risk stratification tools beyond CHA₂DS₂-VA in the very elderly to select the right patients for OACs. Finally, over time, we observed a decrease in stroke and fatal stroke, but did not see the same improvement in stroke-free survival in the very elderly patients. Competing risk of death should, therefore, be considered in this patient group in daily care, and it should be noted that the risk of death was much higher than the risk of a fatal stroke.

Strengths and limitations

The study design offers several benefits, including a large study population, low risk of selection bias, and the opportunity to evaluate multiple

outcomes in the same study cohort over two decades. The ICD-10 code (I48) used to define the AF study population has undergone prior validation, resulting in a positive predictive value (PPV) of 92.6% (95% CI: 88.8%–95.2%).⁴³ Furthermore, we used validated diagnoses for ischaemic stroke (I63, I64, and G45) with PPV of 97%–100%, 57.4%–63.8%, and 57.9%–68.4% respectively.²⁵ Despite the high-quality data in the Danish health registries, certain limitations should be noted. We were unable to adjust our analyses for potential confounders such as body mass index, smoking status, diet, daily activity level, type of AF, and adherence to treatment, as the registries did not contain such information. Moreover, causes of deaths in the registers were not all well validated, and evaluation of cardiovascular mortality was not performed due to the risk of misclassification. The reader should also recognize that concomitant medication was established from prescriptions claimed prior to inclusion. This could cause treatment stopped at inclusion to be carried over, which could introduce bias. Follow-up was limited in the most recent calendar year group due to the end of study date on 31 December 2022. Similarly, comparisons between the different DOACs according to calendar year groups were not possible as the DOACs were introduced during the study period. We used major bleedings to evaluate the bleeding risk; however, validation studies on major bleeding are limited, meaning that we do not know the extent of misclassification.^{26,44} No causal relationships can, therefore, be made, and further investigation of the associations found in this study is needed.

Clinical perspective

This is the first study conducted on the elderly and very elderly AF patients to evaluate changes in the stroke-free survival and incidence of bleeding during a period where OACs were implemented, and aspirin was withdrawn. We demonstrated that the very elderly patients experienced minor improvements in the 5-year stroke-free survival, despite almost full implementation of OACs from 2010 to 2022. In addition, our results suggest that the very elderly patients experienced more ICH. This calls for an increased awareness of risk stratification, to identify the eligible very elderly AF patients, who will benefit from OAC treatment. RCTs on the very elderly AF patients are needed to clarify the exact safety and efficacy of OACs. Moreover, head-to-head comparisons of different DOACs in the very elderly are also an intriguing perspective. Finally, risk evaluation among patients with prior bleeding and stroke events and evaluation of changes in risk factors over time, likewise, remains to be explored.

Conclusion

Following the update of the guidelines for stroke prevention in AF in 2010, the implementation of OAC has been successful in patients aged above 65 years, where 71%–79% receive therapy within 3 months from diagnosis of AF. In the older adult patients and in the elderly patients, stroke-free survival improved by 10.1% and 12.8%, respectively, from 1995 to 2022, without an increased risk of bleeding, and in the very elderly patients, the stroke-free survival improved by 3.5% along with an increased risk of ICH. Therefore, risk stratification beyond CHA₂DS₂-VA is needed to identify patients who may not benefit from OAC therapy.

Supplementary Data

Supplementary data are available at *European Heart Journal* online.

Declarations

Disclosure of Interest

C.B., M.E., E.K.B., M.A., C.S., C.H.G., E.F., A.H., : none. S.L. and N.A.: Unrestricted research grants from Department of Cardiology Herlev-Gentofte Hospital. N.N.: Dagmar Marshalls Fond, Danish Cardiovascular Academy, Christian og Otilia Brorsons rejslegat (12038-2), Reinholdt W Jorck (23-JU-0940), Snedkermester Sophus Jacobsen og hustru Astrid Jacobsens Fond, Murermeister Lauritz Peter Christensen og hustru Kirsten Sigrid Christensens Fond. Speaker fees from AstraZeneca and Bayer. L.Ø.: Independent research grant from the Novo Nordisk Foundation for the study of mitral regurgitation. M.L.: Speaker fees from AstraZeneca and Boehringer Ingelheim. D.M.C.: Independent research grant from the Lundbeck Foundation. L.K.: Speakers honorarium from Astra Zeneca, Boehringer, Bayer, Novartis, and Novo. J.B.O.: Speaker honorarium from Bayer and Organon. M.S.: lecture fees from Novo Nordisk, Astra Zeneca, Boehringer and Novartis. Travel grant to ESC Congress from Bayer.

Funding

This study was funded by an unrestricted hospital grant from the Department of Cardiology, Herlev-Gentofte Hospital.

Data Availability

The data underlying this article are available in the article and in its online [supplementary material](#). Datasets are not publicly accessible, and they were used under specific permissions and licencing. Interested parties may request access to the data through Statistics Denmark (third party).

Ethical Approval

Ethical approval was not required.

Pre-registered Clinical Trial Number

Not applicable.

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